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A Novel Contact Lens Sensor System for Continuous Intraocular Pressure Monitoring: Evaluation of Accuracy in Human Eyes

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Purpose: To evaluate the accuracy of intraocular pressure (IOP) monitoring by a novel contact lens sensor system (CLS) in human eyes.

Design: Cross sectional study.

Participants: Eighty eyes of 80 participants were recruited and divided into 3 groups: (1) 40 normal eyes; (2) 30 eyes with primary open-angle glaucoma (POAG) or ocular hypertension (OHT) with normal IOP (<21 mmHg), including 27 POAG eyes and 3 OHT eyes; and (3) 10 POAG/OHT eyes with high IOP (≥21 mmHg), comprising 4 POAG eyes and 6 OHT eyes.

Methods: Participants wore the CLS to enable continuous monitoring of IOP while they assumed both seated and supine positions, with each position maintained for 10 minutes. Intraocular pressure was also measured by the Goldmann applanation tonometer (GAT) while participants were seated and the Perkins applanation tonometer (PAT) in supine, both before and after CLS wear.

Main Outcome Measures: The average IOP measured by CLS during the final 1-minute of seated and supine positions was compared with IOP measured by GAT and PAT before and after CLS wear. Also, intraclass correlation coefficient and Bland–Altman analyses were performed.

Results: No significant differences were found between pre-CLS GAT and CLS in normal eyes or between all comparisons in POAG/OHT eyes with high IOP ($P > 0.5$). Contact lens sensor system IOP was higher than pre-CLS PAT and post-CLS GAT/PAT IOP in normal eyes ($P < 0.01$), and higher than pre-CLS GAT and post-CLS PAT in POAG/OHT eyes with normal IOP ($P < 0.05$). All IOP differences were within ± 2 mmHg. Intraclass correlation coefficient showed moderate to very strong consistency ($0.51 \leq r \leq 0.95$, $P < 0.05$) except for that between sitting CLS and post-CLS GAT in POAG/OHT eyes with high IOP. Bland–Altman analysis showed that over 80% of points were within ± 5 mmHg and over 60% within ± 3 mmHg.

Conclusions: With good agreement in IOP measurement compared with applanation tonometry in seated and supine positions, across normal and POAG/OHT eyes, the CLS can be used for fairly accurate continuous IOP monitoring.

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Glaucoma is a group of diseases characterized by typical optic atrophy and visual field defects. Increased intraocular pressure (IOP) is currently the most commonly identified and the only verified modifiable risk factor.[1,2] Accurate IOP measurement is crucial for the diagnosis, treatment, and follow-up monitoring of glaucoma. However, IOP is dynamic rather than static, with both short-term and long-term fluctuations.[3–5] A single IOP measurement during office hours can miss IOP peaks in over 60% of glaucoma cases.[6] Even multiple 24-hour IOP measurements, usually conducted 8 to 12 times daily, may not offer complete information.[6,7] Additionally, clinically available tonometers require fixed body positions during examinations and disturb subjects' daily activities, such as sleep. Therefore, continuous 24-hour IOP monitoring becomes an optimal method for comprehensively capturing IOP fluctuations.[6,8,9]

Continuous IOP measurement is primarily enabled by implantable tonometry and contact lens tonometry, with the latter offering the advantage in terms of noninvasion.[9] Despite the development of contact lens sensors in recent years, only a few have been tested in several human eyes.[10–13] Triggerfish (Sensimed AG), the sole commercially available wireless contact lens sensor, has been assessed for the safety and efficacy in 24-hour IOP monitoring.[14–18] However, the data generated by the Triggerfish are presented in millivolt equivalent, posing a challenge in converting electronic signals into IOP values in millimeters of mercury.[18]

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A novel contact lens sensor IOP measurement system (CLS) was developed by a research team from the Hong Kong University of Science and Technology to provide continuous IOP monitoring and to directly output IOP values in millimeters of mercury. Earlier studies showed that this sensor can accurately track pressure fluctuation in a silicone model eye,[19] and tests on enucleated porcine eyes further confirmed the high linearity and sensitivity of the CLS in IOP monitoring.[20] Safety and tolerance assessments, involving continuous 3- and 24-hour wear in adult eyes, have been deemed acceptable.[21] However, evaluation of the accuracy of the CLS for continuous IOP monitoring in human eyes is a crucial step toward the introduction into clinical practice.

Given that the most common body positions in daily life include sitting and supine, accurate IOP monitoring in these 2 postures is necessary. Therefore, this study utilizes the CLS to monitor IOP in healthy participants, patients with primary open-angle glaucoma (POAG), and ocular hypertension (OHT) participants, while assuming seated and supine body positions. Because of the larger fluctuation of IOP for a prolonged period of CLS wear time, the CLS IOP monitoring was only performed over a brief period. This study aimed to evaluate the accuracy of IOP measurement of the CLS in human eyes by agreement analysis of IOP values measured between the CLS and applanation tonometry.

Methods

This study received approval (3031101100) from the Institutional Review Board of the Zhongshan Ophthalmic Center, Sun Yat-sen University, and was conducted in accordance with the Declaration of Helsinki. Ethical approval was obtained from the Ethics Committee of the Zhongshan Ophthalmic Center, Sun Yat-sen University. Oral and written informed consent was obtained from all participants before all procedures.

Study Participants

This cross sectional study enrolled 80 eyes of 80 participants, including 40 healthy individuals, 31 patients diagnosed with POAG, and 9 individuals diagnosed with OHT. Participants were recruited from the glaucoma department at the Zhongshan Ophthalmic Center, Sun Yat-sen University. During recruitment, younger patients were prioritized because of their higher acceptance and better coordination with new devices, particularly those involving contact lens techniques. All participants included in this study were of Chinese ethnicity. Only 1 eye per participant was included. The selected eye was chosen based on the corneal curvature measurement falling within the range of 40 to 44 diopters (D). In cases where both eyes were eligible, the participants were asked for their preference, or the test eye was selected randomly. Eyes were categorized into 3 groups based on diagnosis and IOP measured before CLS wear using the Goldmann applanation tonometer (GAT; Goldmann D-KAT, Keeler): (1) normal eyes group, comprising 40 normal eyes from 40 normal participants; (2) 30 POAG/OHT eyes with normal pre-CLS GAT IOP (<21 mmHg), including 27 eyes from patients with POAG and 3 eyes from OHT participants; and (3) 10 POAG/OHT eyes with high pre-CLS GAT IOP (≥21 mmHg), including 4 POAG eyes and 6 OHT eyes. All POAG and OHT participants received ocular medications or had undergone selective laser trabeculoplasty. Those who were already taking ocular medications continued their regular treatment regimen throughout the trial.

The inclusion criteria of POAG and OHT participants adhered to the European Glaucoma Society Terminology and Guidelines for Glaucoma (5th Edition).[22] Exclusion criteria included (1) individuals <18 years old; (2) presence of ocular surface diseases such as infection and severe dry eyes; (3) history of laser refractive or intraocular surgery; and (4) pregnancy or lactation.

Contact Lens Sensor IOP Measurement System

Building upon the verified correlation between the IOP variations and changes in corneal curvature,[15] the CLS used in this study consisted of a contact lens embedded with IOP sensors, an antenna, a vector network analyzer (Copper Mountain Technologies), a 5-m cable, and the software installed on a laptop (Fig 1A).

The sensor embedded in the contact lens utilizes inductive sensing elements that form inductor–capacitor resonators (resonant circuit) with fixed capacitors, to detect changes in corneal deformation.[19] Multiple resonators serve to reduce sensor noise and mitigate errors caused by variable in vivo conditions.[23] When encapsulated with the medical-grade silicone elastomer (NuSil MED-6015, Avantor) contact lens, the sensor conforms closely to the front surface of cornea for precise tracking of corneal deformations resulting from IOP variations. The sensor modules are made of flexible materials, and the contact lens is available in 3 base curve radii (8.3, 8.6, and 8.9 mm) and 2 diameters (13.8 and 14.2 mm), ensuring suitability for a range of corneal shapes.[21] The base curve radii of 8.3, 8.6, and 8.9 mm were fitted to eyes with corneal curvatures of 43 to 44, 42 to 43, and 40 to 42 D, respectively. The contact lens with diameter of 13.8 mm was chosen initially. If the lens did not fully cover the cornea or extended <0.5 mm beyond the limbus, it was replaced with a larger lens of 14.2 mm in diameter. The central thickness of the contact lens is around 230 μm, and their oxygen permeability (Dk) is 359.8×10^{-11} $(cm^2/s)(mLO_2/[mL \cdot hPa])$.

The inductor–capacitor resonators within the CLS are wirelessly integrated using a ring antenna connected to the reader device. Positioned on the periocular orbit, the antenna facilitates wireless communication with the sensor. The vector network analyzer excites the sensor and interprets the signal continuously. The software installed on a laptop serves as the interface for controlling the reader and managing data acquisition and storage (Fig 1C). During the continuous IOP monitoring by the CLS, participants can perform relatively unconstrained daily activities in various body positions such as sitting and supine.

A signal-IOP model was used to convert the electric signals into IOP values in millimeters of mercury, which was previously developed to reduce IOP error resulting from ocular environmental factors.[23] Positional transitions were performed to optimize the signal-IOP conversion. Both before and during CLS wear, participants assumed positional changes: sitting—horizontal supine—10° head-down tilt (HDT) lying position, with each position maintained for 10 minutes (Fig 2A). Before CLS wear, IOP measurements were obtained using GAT while sitting, and Perkins applanation tonometer (PAT; Perkins MK3, Clement Clarke) while in supine and HDT positions. These baseline IOP measurements by GAT/PAT were fitted into the signal-conversion model when participants repeated the positional transitions during CLS wear (CLS calibration in Fig 2A). After that, an individualized

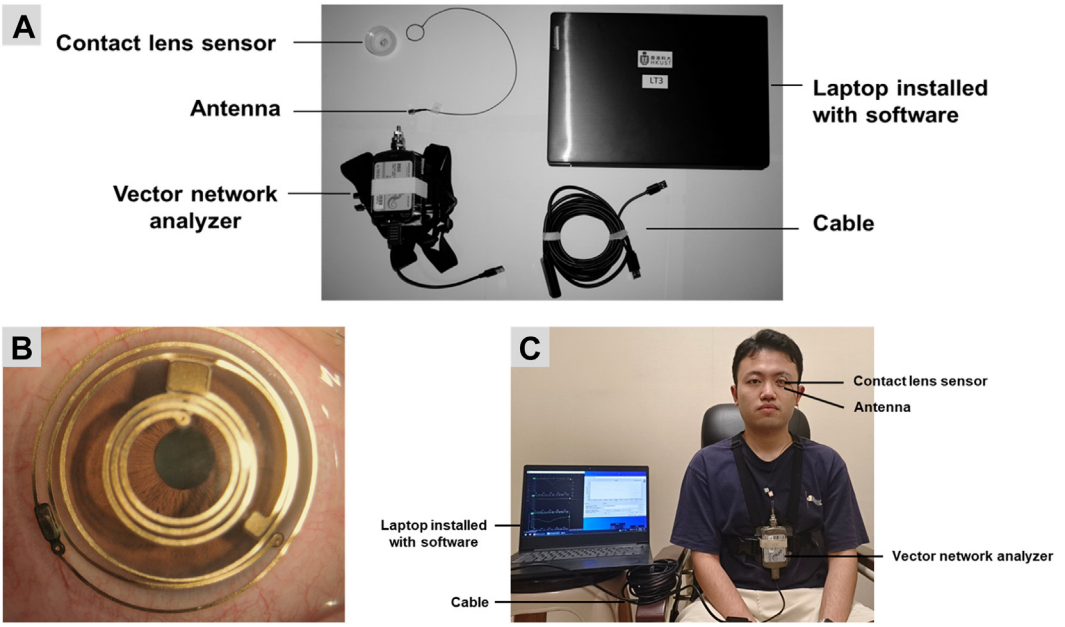


Figure 1. Images of the CLS and IOP measurement. **A,** The constitute of the CLS. **B,** Contact lens in an eye. **C,** Participant wearing the CLS. CLS = contact lens sensor system; IOP = intraocular pressure.

signal-IOP conversion model was established and enabled continuous IOP output. The sensor operates at a sampling frequency of 3 Hz, and the model processes these measurements to determine a stable IOP value. Finally, the CLS provides an IOP measurement at a frequency of 1 Hz.

Applanation Tonometer

Because sitting and supine positions are most common postures in daily life, 2 applanation tonometers, the GAT and PAT, were used to measure IOP in seated and supine positions, respectively. The

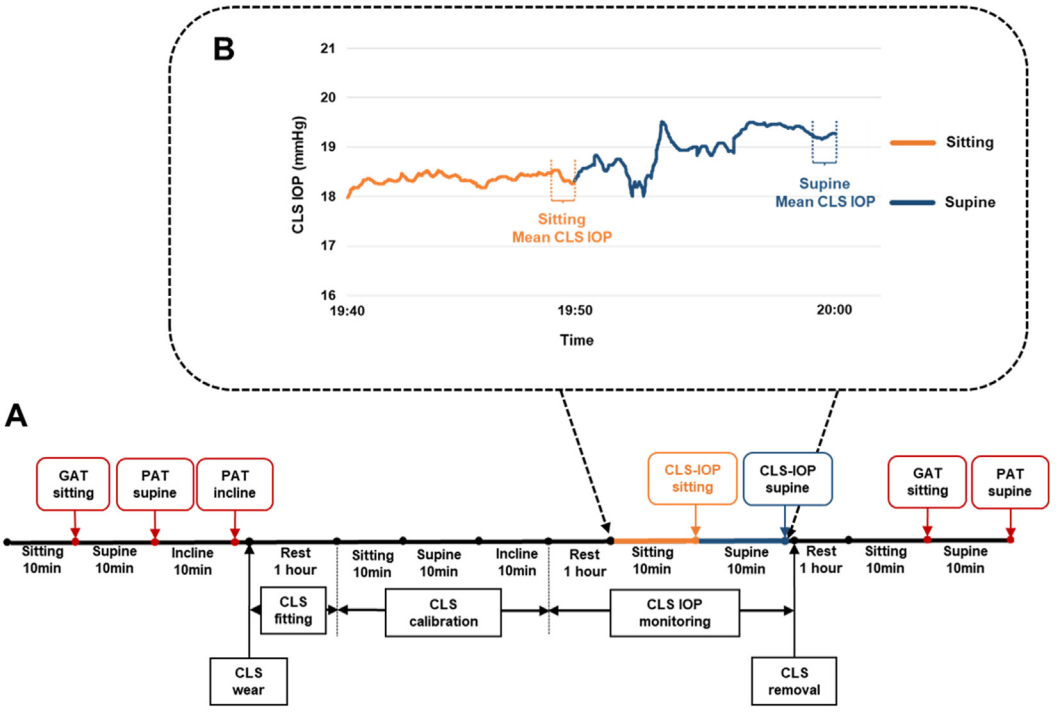


Figure 2. The flow chart of the trial and IOP was continuously monitored by CLS. Participants receiving IOP measurement using applanation tonometers and the CLS. **A,** The flow chart of participants receiving IOP measurements using applanation tonometers and the CLS. **B,** Continuous IOP curve of a participant measured by CLS and mean IOP extracted from the curve for comparative analysis. CLS = contact lens sensor system; GAT = Goldmann applanation tonometer; IOP = intraocular pressure; PAT = Perkins applanation tonometer.

GAT is currently considered the golden standard for IOP measurement, known for its high accuracy and minimal error.[24] The GAT IOP measurement was performed at the slit lamp microscope with the participant in the seated position. The PAT, a handheld applanation tonometer, was used to measure IOP in participants while they were in both supine and HDT positions. All IOP measurements using GAT and PAT were conducted by a single trained technician, who had over a year of training and demonstrated high accuracy and repeatability in IOP measurements. To enhance measurement reliability, if the first 2 readings differed by ≤1 mmHg, their average was recorded. If the difference between the first 2 readings was >1 mmHg, a third reading was taken and averaged with the first 2 readings.

Description of Trials

All participants first underwent ophthalmic examinations, including best corrective visual acuity, slit lamp biomicroscopy, and corneal radius (swept-source OCT, CASIA SS-1000, Tomey Corporation). Next, participants were required to perform positional transitions with IOP measured by applanation tonometry (Fig 2A). After this, participants wore the CLS for approximately 3 hours, including the CLS fitting period, the IOP calibration period, and the IOP monitoring period. During the CLS fitting period, participants were instructed to maintain a seated position and avoid any vigorous activity for 1 hour. Participants were instructed to close their eyes gently for 20 minutes for lens fitting and were then allowed to keep their eyes open while maintaining a relaxed state. To minimize potential influences on IOP and IOP measurements, no near-vision tasks, food, or fluid intake were permitted throughout the trial. This duration was chosen to allow for lens settling, achieve a stable signal, and allow the IOP to return to baseline. During the IOP calibration period, where IOP was monitored using the CLS in seated, supine, and HDT postures, with each position maintained for 10 minutes, an individual signal-IOP conversion model was developed. Then participants had a sitting period of 1 hour to allow IOP to return to their baseline again. Participants were then directed to sit and subsequently lie down for 10 minutes. The average of continuous CLS IOP values over 1 minute at the end of each position period was extracted for analysis (Fig 2B). Finally, after 1 hour of seated rest after the CLS removal, IOP was measured by GAT and PAT at the end of 10 minutes of performing sitting and supine positions, respectively. All study procedures were carried out on a single day, with IOP measurements performed from the afternoon to the early evening hours.

Statistical Analysis

Descriptive statistics were performed to calculate the clinical characteristics of study participants. Data are presented as mean values ± standard deviation along with the 95% confidence interval. Intraocular pressure values were compared using repeated measures analysis of variance and post hoc *t* tests with Bonferroni correction. Intraclass correlation and Bland–Altman analysis were applied to assess the agreement between the IOP values measured by CLS and GAT/PAT. Intraclass correlation coefficient (ICC), ranging from 0 to 1, was interpreted as follows: ICC <0.2 indicates poor consistency, 0.2 to 0.4 indicates general consistency, 0.4 to 0.6 indicates moderate consistency, 0.6 to 0.8 indicates strong consistency, and 0.8 to 1.0 indicates very strong consistency.[25] For all tests, a value of $P < 0.05$ was considered statistically significant. All statistical analyses were performed with SPSS version 22.0 (IBM SPSS).

Results

Participants Clinical Characteristics

Eighty participants were enrolled in the study. Among them, 41 were men, and 39 were women, with a mean age of 29.7 ± 11.3 years. All participants completed the continuous IOP monitoring using the CLS and applanation tonometry on 1 eye. The characteristics of participants in each group are summarized in Table 1.

Table 1. Characteristics of Participants in Each Group

Characteristics	Normal Group (n = 40)	Patient Group with Pre-CLS GAT IOP <21 mmHg (n = 30)	Patient Group with Pre-CLS GAT IOP ≥21 mmHg (n = 10)
POAG (n)	-	27	4
OHT (n)	-	3	6
Age (yrs)	23.8 ± 3.2	35.0 ± 14.8	37.1 ± 9.1
Sex (male/female)	17/23	18/12	6/4
Eye (right/left)	26/14	19/11	6/4
Treatment (n)			
Prostaglandin analogs	-	14	8
Beta-receptor antagonists	-	7	1
Alpha-2 adrenergic agonists	-	3	0
Carbonic anhydrase inhibitors	-	2	0
Selective laser trabeculoplasty	-	4	1
Lens type			
Base curve: 8.3 mm, diameter: 13.8 mm	6	0	0
Base curve: 8.6 mm, diameter: 13.8 mm	18	14	4
Base curve: 8.9 mm, diameter: 13.8 mm	6	3	0
Base curve: 8.3 mm, diameter: 14.2 mm	3	2	1
Base curve: 8.6 mm, diameter: 14.2 mm	7	10	5
Base curve: 8.9 mm, diameter: 14.2 mm	0	1	0

CLS = contact lens sensor system; GAT = Goldmann applanation tonometer; IOP = intraocular pressure; OHT = ocular hypertension; POAG = primary open-angle glaucoma.

Table 2. Intraocular Pressure Values Measured by CLS, Pre-CLS, and Post-CLS GAT/PAT

Groups	N	Position	Pre-CLS GAT/PAT (mmHg)		CLS (mmHg)		Post-CLS GAT/PAT (mmHg)	
			Mean ± SD	*95% CI*	*Mean ± SD*	*95% CI*	*Mean ± SD*	*95% CI*
Normal	40	Sitting	12.7 ± 2.0	12.1–13.4	12.8 ± 2.6	12.0–13.6	11.7 ± 1.9	11.0–12.3
		Supine	13.9 ± 2.4	13.1–14.6	14.8 ± 3.1	13.9–15.8	13.6 ± 2.8	12.7–14.5
Patient (pre-GAT IOP <21 mmHg)	30	Sitting	16.2 ± 3.7	14.9–17.6	17.3 ± 3.8	15.9–18.7	16.1 ± 3.4	14.9–17.4
		Supine	19.8 ± 4.3	18.2–21.4	20.4 ± 4.0	19.0–22.0	19.0 ± 3.8	17.6–20.4
Patient (pre-GAT IOP ≥21 mmHg)	10	Sitting	24.4 ± 2.1	22.9–25.9	23.6 ± 3.0	21.5–25.7	25.5 ± 7.3	20.3–30.7
		Supine	24.5 ± 3.2	22.2–26.8	25.0 ± 4.0	22.1–27.9	25.5 ± 4.7	22.2–28.8

CI = confidence interval; CLS = contact lens sensor system; GAT = Goldmann applanation tonometer; IOP = intraocular pressure; PAT = Perkins applanation tonometer; SD = standard deviation.

IOP Values Measured by CLS and GAT/PAT

The IOP measured by CLS and GAT/PAT in each group are presented in Table 2. The mean CLS IOP in normal participants was 12.8 ± 2.6 and 14.8 ± 3.1 mmHg in sitting and supine, respectively, with applanation IOP in corresponding positions being 12.7 ± 2.0 and 13.9 ± 2.4 mmHg before CLS, and 11.7 ± 1.9 and 13.6 ± 2.8 mmHg after CLS monitoring. The POAG/OHT eyes with normal IOP had a mean CLS IOP of 17.3 ± 3.8 mmHg in sitting and 20.4 ± 4.0 mmHg in supine, with applanation IOP of 16.2 ± 3.7 and 19.8 ± 4.3 mmHg before CLS, and 16.1 ± 3.4 and 19.0 ± 3.8 mmHg after CLS wear in corresponding positions. For POAG/OHT eyes with high IOP, the mean CLS IOP in sitting and supine was 23.6 ± 3.0 and 25.0 ± 4.0 mmHg, respectively. The applanation IOP of them in sitting and supine was 24.4 ± 2.1 and 24.5 ± 3.2 mmHg before CLS monitoring and 25.5 ± 7.3 and 25.5 ± 4.7 mmHg after CLS wear, respectively.

Comparison of IOP Values Measured by CLS and GAT/PAT

Comparisons among IOP values measured by CLS and applanation tonometers showed no significant difference between pre-CLS GAT IOP and CLS IOP in the normal group ($P > 0.999$), as well as in every comparison within the POAG/OHT eyes with high IOP (all $P > 0.5$). However, CLS IOP was significantly higher than pre-CLS PAT IOP (mean difference = 1.0 mmHg, $P = 0.005$) and post-CLS GAT/PAT IOP (mean difference = 1.2 mmHg, $P = 0.008$; mean difference = 1.2 mmHg, $P = 0.004$) in normal participants. In POAG/OHT eyes with normal IOP, the CLS IOP was significantly higher than pre-CLS GAT (mean difference = 1.0 mmHg, $P = 0.031$) and post-CLS PAT (mean difference = 1.5 mmHg, $P < 0.001$). Mean IOP differences between CLS and applanation tonometer were consistently <2 mmHg in all 3 groups (Fig 3). Additionally, the post-CLS GAT IOP was lower than pre-CLS GAT IOP in normal eyes (mean difference = 1.0 mmHg, $P = 0.010$), and the post-CLS PAT IOP was also lower than the prior PAT measurement in POAG/OHT eyes with normal IOP (mean difference = 0.8 mmHg, $P = 0.010$).

Agreement Evaluation between IOP Values Measured by CLS and GAT/PAT

The correlation of IOP measurements between CLS and PAT/GAT in 3 groups is illustrated in Figure 4. The ICC showed good consistency between IOP measured by CLS and applanation devices ($0.51 \leq r \leq 0.95$, all $P < 0.05$), except for that between sitting CLS and post-CLS GAT ($r = 0.32$, $P = 0.165$) in POAG/OHT eyes with high IOP.

The agreement of IOP measurements between CLS and GAT/PAT is depicted in Figure 5. The percentage of IOP differences within 5 mmHg was 95%, 93%, and 90% between pre-GAT and CLS; 97%, 90%, and 80% between post-GAT and CLS; 97%, 96%, and 100% between pre-PAT and CLS; 93%, 96%, and 100% between post-PAT and CLS, in normal eyes, POAG/OHT eyes with normal IOP, and POAG/OHT eyes with high IOP, respectively. The percentage of IOP differences within 3 mmHg was 90%, 86.67%, and 90% between pre-GAT and CLS; 77.5%, 83.33%, and 60% between post-GAT and CLS; 92.5%, 90%, and 100% between pre-PAT and CLS; 85%, 93.3%, and 100% between post-PAT and CLS, in normal eyes, POAG/OHT eyes with normal IOP, and POAG/OHT eyes with high IOP, respectively.

Discussion

In clinical practice, 24-hour IOP measurements are vital for glaucoma diagnosis and assessment of IOP variations before and after treatments, particularly in cases of disease progression. Continuous IOP monitoring via contact lens tonometry offers advantages such as increased measurement frequencies and fewer disruptions to the natural physiological state. This study introduces a new device enabling continuous IOP monitoring in human eyes and compares results of the CLS with applanation tonometry, in both sitting and supine positions, among normal participants and POAG/OHT individuals. Results revealed slight differences, significant correlations, and fair agreement between the CLS and applanation tonometry.

The average difference of IOP values between the CLS and applanation tonometry was approximately within ± 2

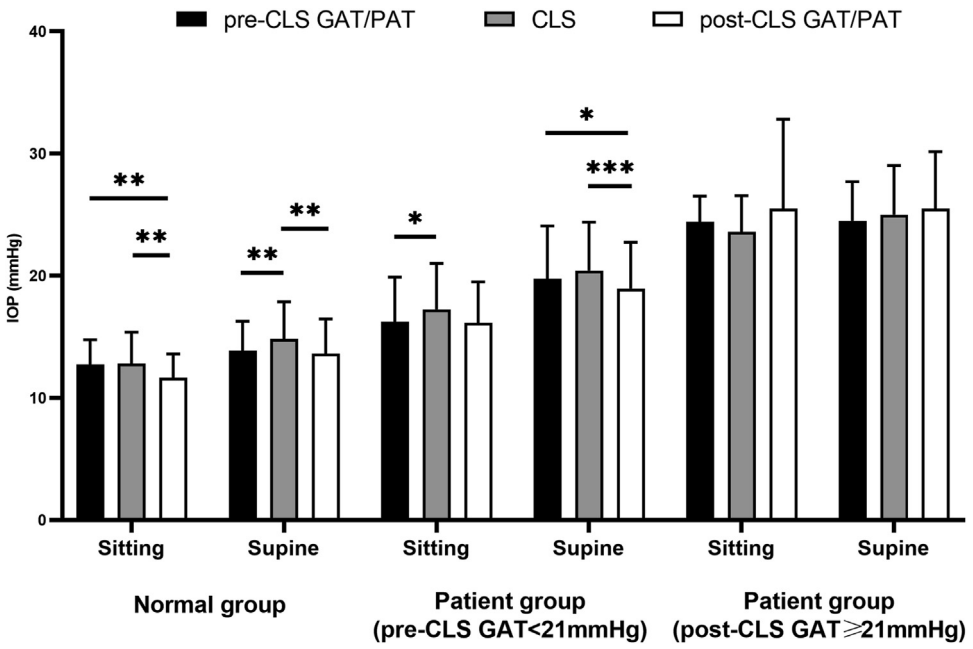


Figure 3. Comparisons of IOP values measured by CLS, pre-CLS, and post-CLS GAT/PAT. CLS = contact lens sensor system; GAT = Goldmann applanation tonometer; IOP = intraocular pressure; PAT = Perkins applanation tonometer. *$P < 0.05$; **$P < 0.01$; ***$P < 0.001$.

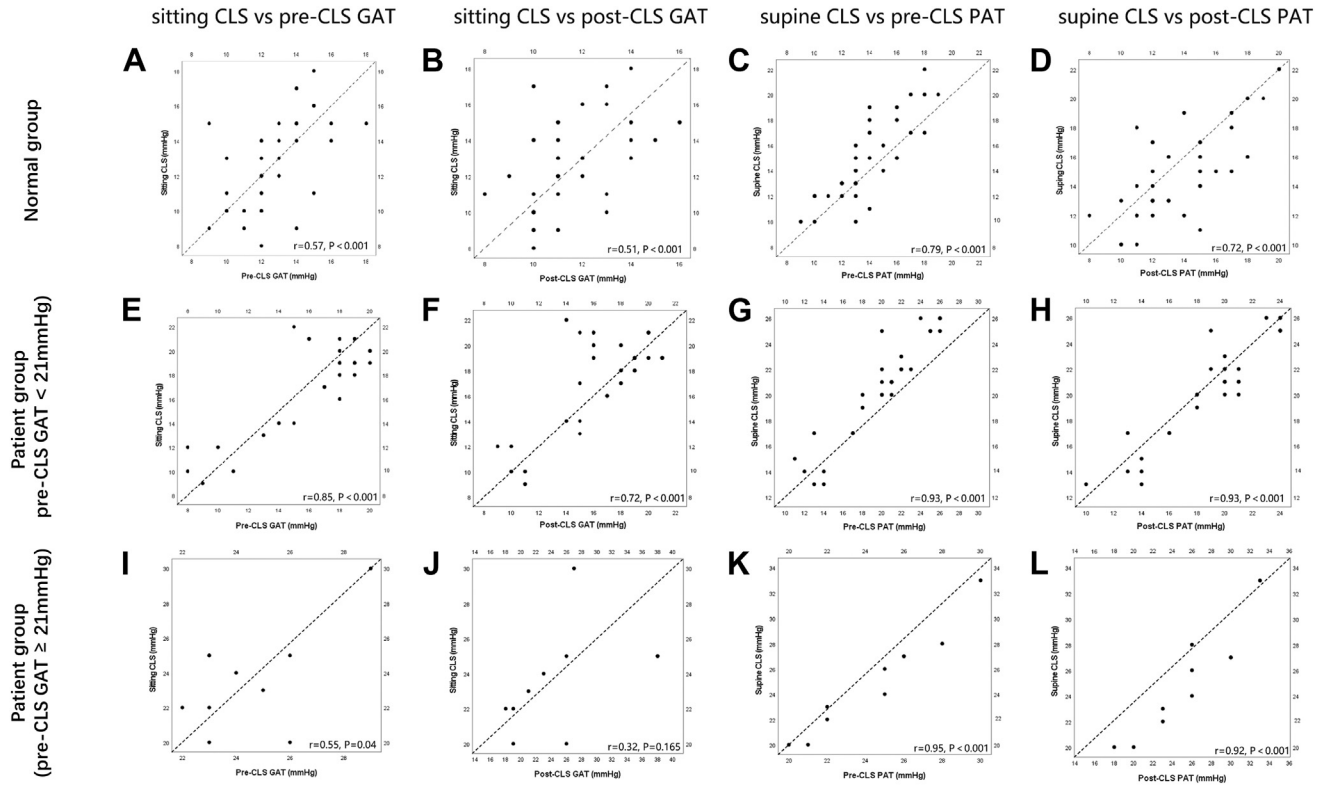


Figure 4. Correlation of IOP values measured between CLS and applanation tonometry. **A,** sitting CLS IOP versus pre-CLS GAT IOP in normal group. **B,** sitting CLS IOP versus post-CLS GAT IOP in normal group. **C, D,** supine CLS IOP versus pre- and post-CLS PAT IOP in normal group. **E, F,** sitting CLS IOP versus pre- and post-CLS GAT IOP in patient group with normal IOP. **G, H,** supine CLS IOP versus pre- and post-CLS PAT IOP in patient group with normal IOP. **I, J,** sitting CLS IOP versus pre- and post-CLS GAT IOP in patient group with high IOP. **K, L,** supine CLS IOP versus pre- and post-CLS PAT IOP in patient group with high IOP. CLS = contact lens sensor system; GAT = Goldmann applanation tonometer; IOP = intraocular pressure; PAT = Perkins applanation tonometer.

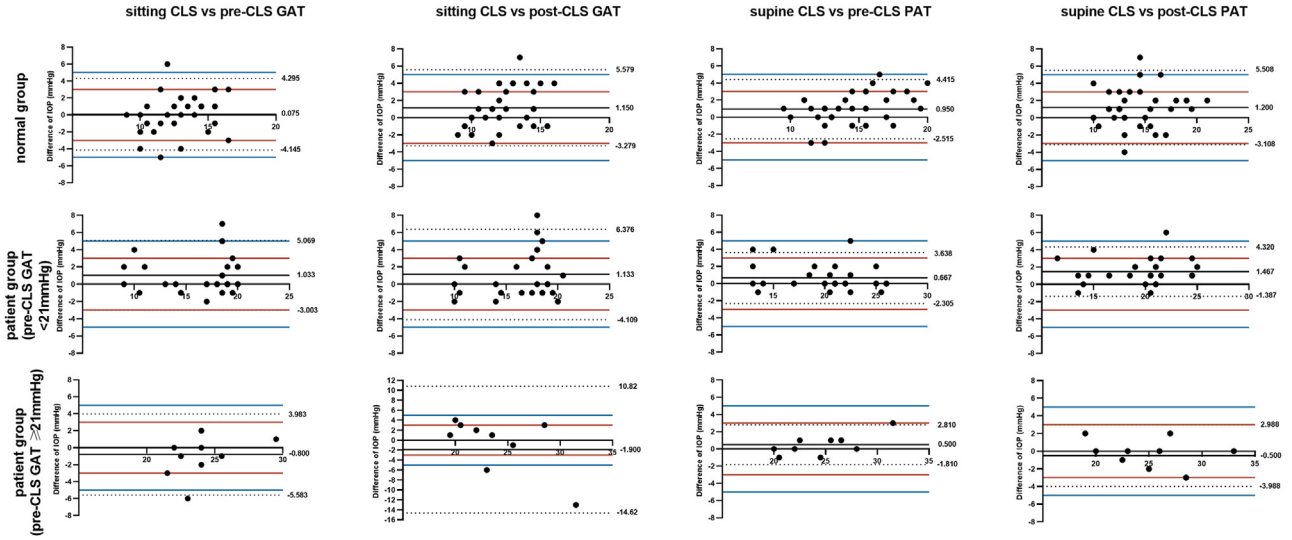


Figure 5. Bland–Altman plot of means against the difference between the IOP values measured by CLS and applanation tonometry. Bland–Altman plot comparing the CLS and applanation tonometry. The x-axis represents the mean of the 2 methods, and the y-axis represents the difference between the 2 methods (CLS–applanation tonometry). The solid line represents the mean difference, whereas the dashed lines represent the 95% limits of agreement (mean difference ± 1.96 standard deviation). The red lines represent the limits of agreement at ±3 mmHg, and the blue lines represent ±5 mmHg. CLS = contact lens sensor system; GAT = Goldmann applanation tonometer; IOP = intraocular pressure; PAT = Perkins applanation tonometer.

mmHg, in both seated and supine positions in all 3 groups, with or without statistical significance. Some of the differences may result from the spontaneous IOP variations with time[26] because the entire trial process lasted about 5 hours. Also, although participants were asked to perform 1-hour seated rest before positional transitions to allow the IOP to revert to baseline (IOP in sitting measured by GAT before CLS wear), the influence of prior body positions on IOP may not have been completely elucidated. This may partly explain why CLS IOP tended to be higher than the pre-CLS GAT/PAT IOP in most comparisons, with or without statistical significance. The CLS IOP was higher than the post-CLS GAT/PAT IOP in normal eyes and POAG/OHT eyes with normal IOP. The observed IOP differences could also be attributed to diurnal fluctuations. Also, the post-CLS applanation IOP was lower than that measured before CLS wear in some groups. It is speculated that wearing the CLS might affect applanation tonometer measurements after the contact lens is removed, as the anterior surface of the cornea may flatten due to the tight fit. This could result in a smaller corneal curvature and lower IOP values observed by the applanation tonometer. Undeniably, the CLS can have some measurement errors. Intraocular pressure measurements can vary widely (from 0.4 to 3.4 mmHg) dependent upon the tonometer used.[27,28] For the Pressure-Measuring Contact Lens, another contact lens sensor technique that can output IOP in millimeters of mercury, the average difference of IOP measured between it and GAT was from 0.1 to 6.5 mmHg in human eyes.[11,12] In summary, the difference between CLS in this study and applanation tonometry was deemed modest and acceptable.

The Bland–Altman plot is a commonly used method to analyze the agreement between different measurement devices or techniques. The ± 3 mmHg threshold is generally accepted as the limit for testing new tonometers. In the current study, >80% of CLS measurements fell within ± 3 mmHg of IOP values measured by applanation tonometers in all comparisons, except for that between CLS IOP and post-CLS GAT IOP in normal eyes and POAG/OHT eyes with high IOP. This proportion was comparable to those reported among commonly used tonometers (from 51.2% to 84%).[29–31] Also, >80% of CLS measurements were within ± 5 mmHg of IOP measured with an applanation tonometer, which was also comparable to those reported among CLS-resembling techniques (from 62.5% to 97.2%).[10,12,27,28] Moreover, most ICC analyses showed a moderate to very strong correlation between IOP measured by CLS and applanation tonometry, which was similar to previous reports among multiple tonometers.[27,32] However, the correlation was weak, and the percentage of IOP reading difference within 3 mmHg was only 60% between sitting CLS and post-CLS GAT in POAG/OHT eyes with high IOP. This may be attributed to the small sample size and the possible spontaneous larger IOP variations in POAG/OHT eyes with high IOP.[33] Nonetheless, the capability of the CLS to provide accurate and stable IOP monitoring in eyes with high IOP needs further verification with larger samples. The results of this study collectively suggest fair agreement between IOP values measured by the CLS and applanation tonometers in both normal participants and POAG/OHT participants.

With a similar working principle of the Triggerfish, the CLS used in this study can capture variations in IOP that lead to changes in corneal shape via embedded sensors. Prior studies have demonstrated high linearity and

sensitivity in IOP monitoring on enucleated porcine eyes for both the Triggerfish and CLS.[20,34] Although the Triggerfish is currently the sole commercially available device, it cannot translate the electric signal into IOP values.[18] The Triggerfish relies on a resistive strain gauge-based sensor, whereas the CLS utilizes an inductor–capacitor based sensor to measure corneal deformation.[19,34] In human eyes, an individual calibration process is used to convert the electric signals to IOP values in millimeters of mercury based on a previously developed model.[23] In the case of the Triggerfish, the encapsulated chip results in a sensor thickness of 583 μm,[35] whereas the chipless sensor of the CLS in this study was thinner. Therefore, the reactance sensing approach holds potential advantages in terms of safety and comfort when compared with the resistive approach involving a rigid chip.

To date, only a few contact lens sensors have been tested in human eyes. However, the sample size of human participants ranged from 6 to 10, and some were conducted only in the eyes of people with normal IOP.[10–13] The accuracy of IOP measurement in POAG/OHT eyes, especially those with high IOP, should be tested and ensured before application in clinical settings. Implantable IOP tonometry was an alternative to continuous tonometry, which provides advantages such as direct IOP sensing and long-time monitoring. However, the shortcomings were prominent such as inevitable invasive implantation operation and possible sensor malfunction.[18,36]

This study has several limitations. First, the sample size of patients with POAG and OHT participants, especially those with high IOP, was relatively small. A more extensive study is required for robust analyses. Although seated and supine positions are the most common positions in daily life, stable and accurate IOP measurement during other activities, including inclining, standing, and some exercises like walking, are still important. In addition, the CLS wearing time in this study was only 3 hours. Verification of the accuracy in continuous monitoring of IOP for up to 24 hours is a crucial prerequisite for future application of the CLS. The individual calibration and availability of multiple contact lens sizes help improve IOP measurement accuracy. However, the performance of the CLS in eyes with corneal curvatures outside the designed range remains uncertain. Because of the relatively small size, a detailed analysis of the correlation between corneal curvature and CLS accuracy was not performed. Future studies should further investigate this relationship and explore adjustments to optimize sensor performance across a wider range of corneal curvatures.

In conclusion, this study serves as proof of the availability of the novel CLS for continuous IOP measurements in both normal participants and POAG/OHT participants across various postures over a short period. With good agreement with applanation tonometry, the CLS offers a promising approach to continuous 24-hour IOP measurement in human eyes.

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Footnotes and Disclosures

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HUMAN SUBJECTS: Human subjects were included in this study. This study received approval (3031101100) from Institutional Review Board of the Zhongshan Ophthalmic Center, Sun Yat-sen University, and was conducted in accordance with the Declaration of Helsinki. Ethical approval was obtained from the Ethics Committee of the Zhongshan Ophthalmic Center, Sun Yat-sen University. Oral and written informed consent was obtained from all participants before all procedures.

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Abbreviations and Acronyms:
CLS = contact lens sensor system; **D** = diopters; **GAT** = Goldmann applanation tonometer; **HDT** = head-down tilt; **IOP** = intraocular pressure; **ICC** = intraclass correlation coefficient; **OHT** = ocular hypertension; **PAT** = Perkins applanation tonometer; **POAG** = primary open-angle glaucoma.

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